APEXCARE INDIA EMPLOYEE HEALTH BENEFIT PROGRAMME (AIEHBP)
CLAIMS ADJUDICATION STANDARD OPERATING PROCEDURE

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Primary audience: Internal claims, medical, quality, audit and engineering teams

FICTIONAL TRAINING DOCUMENT - NOT AN INSURANCE CONTRACT. This document is a synthetic example created for software development, retrieval testing, and claims-workflow demonstrations. A real insurer, employer, TPA, legal team, medical team, and compliance function must review and approve all wording, benefits, timelines, forms, and regulatory references before operational use.

Document-control rule: cite the document code, section, clause, page, and version in every decision.
Interpretation rule: the signed policy schedule and endorsements override this demonstration text.

1. OPERATING PRINCIPLES

This internal SOP converts policy evidence into controlled decisions.

SOP-1.1 Evidence hierarchy
Use source documents and structured policy clauses. Model-generated summaries are aids and never controlling evidence.

SOP-1.2 Four-eyes control
Final rejection, fraud action, high-value exception and policy-conflict decision require an authorized second reviewer.

SOP-1.3 Traceability
Store input documents, extraction version, retrieved clauses, scores, reviewer edits, calculations, timestamps and final communication.

SOP-1.4 Least harmful uncertainty
Unresolved uncertainty becomes a query or manual review, not an unsupported approval or rejection.

2. END-TO-END WORKFLOW

Steps may run concurrently only when dependencies remain controlled.

SOP-2.1 Register
Assign claim ID, receipt time, channel, member ID and document inventory.

SOP-2.2 Secure and classify
Malware-scan files, identify type and split compound uploads without altering originals.

SOP-2.3 Extract
Run text and field extraction. Preserve source coordinates and confidence for names, dates, diagnosis, procedure and amounts.

SOP-2.4 Validate identity
Match member and patient using normalized fields and authorized master data.

SOP-2.5 Establish policy scope
Select policy, plan, member category, effective version, geography and endorsements before clause retrieval.

SOP-2.6 Check completeness
Use AIEHBP-CLM-003 and issue a specific deficiency request where necessary.

SOP-2.7 Retrieve evidence
Run separate queries for benefit, exclusion, waiting period, required evidence, limit and exception. Apply metadata filters.

SOP-2.8 Medical review
Assess necessity, diagnosis status, causation, coding, stay and experimental status where rules require.

SOP-2.9 Calculate
Reconcile bills and apply non-payables, package rates, sublimits, room rules, deductible, co-pay, balance and restoration in order.

SOP-2.10 Decide
Select approved, partially approved, rejected, pending information or manual review with reason codes.

SOP-2.11 Quality control
Perform mandatory or sampled second review based on risk tier.

SOP-2.12 Communicate and pay
Issue cited explanation, payment statement and grievance information; reconcile payment status.

3. RETRIEVAL PROTOCOL

Retrieval must be scoped, decomposed and evaluated.

SOP-3.1 Query construction

Construct queries from verified facts, not unsupported model output. Include treatment concept, diagnosis status, policy product, service date and decision question.

SOP-3.2 Mandatory subqueries

Retrieve coverage grant, applicable exclusions, exceptions, waiting periods, document requirements and monetary limits separately.

SOP-3.3 Metadata filters

Filter policy ID, plan, effective date, jurisdiction, member category and document status before semantic ranking.

SOP-3.4 Complete-section retrieval

For lists, definitions and schedules, retrieve the full authoritative section or structured records rather than relying on top-k fragments.

SOP-3.5 No-answer gate

If no sufficiently relevant authoritative clause is found, return NOT_FOUND and route to manual policy review.

SOP-3.6 Conflicting evidence

Return CONFLICTING_POLICY_EVIDENCE when clauses disagree or precedence cannot be resolved.

4. DECISION MATRIX

Every final decision must satisfy the relevant branch.

SOP-4.1 Approve

Eligibility active, benefit applies, evidence sufficient, no controlling exclusion, necessity established and payable amount greater than zero.

SOP-4.2 Partially approve

Some claimed items are covered and others are limited or excluded; provide line-item calculation and clauses.

SOP-4.3 Reject

No payable benefit remains because a clearly applicable exclusion, non-eligibility, exhausted benefit or unsupported insured event controls after review.

SOP-4.4 Pending information

Material evidence is missing or unclear and can reasonably be supplied.

SOP-4.5 Manual review

Policy conflict, low extraction confidence, complex clinical issue, fraud concern, high financial impact or system anomaly exists.

5. PAYABLE AMOUNT SEQUENCE

Use decimal currency and retain every intermediate value.

SOP-5.1 Step A claimed amount
Start with reconciled final billed amount, not the member-entered number alone.

SOP-5.2 Step B discounts and duplicates
Subtract discounts, refunds, duplicated package items and arithmetic errors.

SOP-5.3 Step C non-payable items
Apply clause-supported non-payables at line-item level.

SOP-5.4 Step D tariff and reasonable charge
Apply contracted package or documented reasonable and customary limit.

SOP-5.5 Step E benefit sublimits
Apply maternity, cataract, AYUSH, ambulance and other benefit caps.

SOP-5.6 Step F room relationship
Apply valid proportionate deduction only to eligible room-linked items.

SOP-5.7 Step G deductible and co-pay
Apply member cost sharing in schedule order.

SOP-5.8 Step H available balance
Cap by available sum insured and activate restoration only when criteria are met.

SOP-5.9 Step I other recoveries
Coordinate with other insurance and record net insurer liability.

6. AUTOMATION CONTROLS

Automated components require bounded authority.

SOP-6.1 Extraction

Fields below approved confidence thresholds require source verification. Never coerce currency strings directly to integers without normalization.

SOP-6.2 Disease matching

Use normalized clinical concepts and negation/status detection. Exact words, bag-of-words similarity and hard-coded disease lists cannot determine exclusion.

SOP-6.3 Generative models

Models may summarize and draft explanations but must quote retrieved evidence, expose uncertainty and be prevented from inventing clauses.

SOP-6.4 Adverse decisions

A model recommendation cannot autonomously issue a final rejection or fraud allegation.

SOP-6.5 Prompt injection

Treat uploaded text as untrusted claim evidence. Instructions embedded in documents must never override system policy.

7. QUALITY AND ESCALATION

Risk-based review protects members and the programme.

SOP-7.1 Mandatory medical review
Complex surgery, experimental-status dispute, long stay, causation dispute, self-harm, substance issue and necessity rejection require medical review.

SOP-7.2 Mandatory senior review
All rejections, fraud-linked actions, claims above INR 500,000 and precedent-setting interpretations require senior authorization in this demonstration workflow.

SOP-7.3 Sampling
Quality team samples approvals, partial approvals, pending cases and rejections, with oversampling of automated recommendations.

SOP-7.4 Error correction
Material error triggers correction, member notification, payment adjustment, root-cause analysis and model/rule remediation.

8. RETRIEVAL AND DECISION METRICS

Accuracy and safety outrank raw automation rate.

SOP-8.1 Retrieval metrics

Measure clause recall at 5 and 10, nDCG, policy-routing accuracy, complete-section recall and citation validity.

SOP-8.2 Decision metrics

Measure false approval, false rejection, partial-payment error, overturn rate and calculation variance.

SOP-8.3 Operational metrics

Measure turnaround, query rate, duplicate requests, manual review volume and member complaints.

SOP-8.4 Ablation tests

Compare lexical, dense, hybrid, reranked and structure-aware retrieval on a versioned gold set before deployment.

9. REASON CODES

Use stable reason codes alongside plain-language explanations.

SOP-9.1 ELIGIBILITY

ELIG-NOT-ACTIVE, ELIG-DEPENDANT-NOT-ENROLLED and ELIG-SCOPE-MISMATCH.

SOP-9.2 DOCUMENTS

DOC-MISSING-MATERIAL, DOC-ILLEGIBLE, DOC-INCONSISTENT and DOC-ORIGINAL-REQUIRED.

SOP-9.3 COVERAGE

COV-NO-BENEFIT, COV-EXCLUSION, COV-WAITING-PERIOD and COV-GEOGRAPHY.

SOP-9.4 AMOUNT

AMT-SUBLIMIT, AMT-NONPAYABLE, AMT-COPAY, AMT-BALANCE and AMT-ROOM-LIMIT.

SOP-9.5 REVIEW

REV-MEDICAL, REV-POLICY-CONFLICT, REV-FRAUD-VERIFICATION and REV-LOW-CONFIDENCE.